TECHNICAL PORTFOLIO • APPLIED DATA SCIENCE & STATISTICAL INFERENCE

Bank Marketing Term Deposit Campaign: EDA, Hypothesis Testing, Bayesian Inference & SQL Insights

An end-to-end analytical study modeling client demographics, campaign exposure dynamics, and telemarketing conversion probability on 45,211 retail bank contacts.

PROJECT DOMAIN	ANALYTICS STACK	VERIFIED COHORT	CONVERSION BENCHMARK
Retail Banking / Direct Marketing	**Python, Pandas, SciPy, SQLite**	**45,211 Records (17 Features)**	**11.70% (5,289 Subscribed)**

1. Introduction & Project Background

Direct telemarketing campaigns remain a cornerstone customer acquisition channel for commercial banking institutions offering term deposit products. In a term deposit contract, funds are placed with the bank for a fixed maturity at a guaranteed interest yield, providing institutions with low-volatility capital for liquidity planning and credit operations. However, untargeted telemarketing incurs significant operational expenses, causes outbound agent fatigue, and accelerates customer churn due to repetitive contact friction.

This project investigates the historical campaign records of a leading Portuguese retail banking institution. By taking the project from raw, uncurated data to an engineered database structure, this report establishes a unified exploratory, statistical, and relational SQL framework. The primary goal is to determine the empirical profiles of responsive clients, isolate high-impact contact mechanisms, statistically validate behavior differences across demographic segments, and formulate prescriptive conversion strategies for institutional resource allocation.

2. Project Description & Core Architecture

Core Objectives & Functionality

- **Multi-Stage Data Engineering:** Ingest raw semi-structured files, treat implicit missing flags ("unknown"), resolve skewed distributions, and derive behavior indices.
- **Empirical Statistical Inference:** Transition beyond visual impressions through two-sample Welch t-tests, Pearson Chi-square independence tests, and two-proportion Z-tests.
- **Bayesian Posterior Induction:** Calibrate prior probabilities with conjugated Beta-Binomial models based on historical campaign outcomes (`poutcome`).

Expected Outcomes & Impact

- **SQL Business Intelligence:** Deploy relational schemas in SQLite to synthesize aggregate KPIs, multi-tier segment matrices, and campaign intensity decay rates.
- **Data-Driven Targeting Framework:** Replace blanket calling runs with a priority index highlighting high-conversion cohorts (students, retirees, previous successes).
- **Efficiency Guardrails:** Define call volume caps (≤3 contacts) to prevent operational diminishing returns.

System Analytical Framework



3. Problem Statement & Business Context

Commercial retail banks face acute margin pressure and escalating customer acquisition costs in competitive deposit markets. In the analyzed campaign, 45,211 outbound calls generated only 5,289 successful deposit subscriptions, yielding an aggregate baseline conversion rate of **11.70%**. This indicates that **88.30%** (39,922 calls) represented unproductive staff hours and wasted operational expenditures.

The underlying organizational problem stems from non-segmented, indiscriminate outreach. Outbound agents lack statistically grounded targeting criteria, leading to call exhaustion over demographic cohorts with low propensity to subscribe, while high-propensity niche segments remain under-leveraged. The technical objective of this project is to develop an empirical intelligence layer that uncovers predictive subscriber traits, models the trade-off between contact frequency and conversion probability, and optimizes marketing ROI.

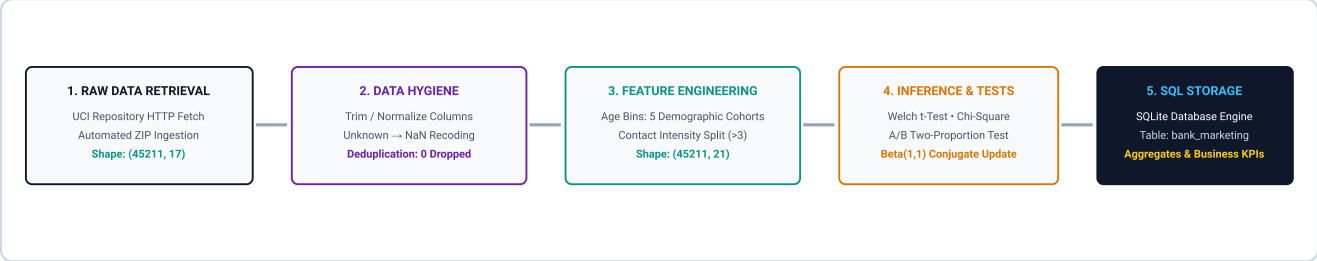
4. Dataset Description, Processing & Lineage

The source dataset was obtained from the UCI Machine Learning Repository (Bank Marketing Dataset), containing 45,211 observation rows and 17 original attribute columns capturing four distinct contextual dimensions: client personal demographics, financial solvency indicators, campaign contact attributes, and previous marketing outcomes.

Feature Category	Variable Names	Data Type / Scale	Analytical Role in Campaign Modeling
Demographics	age, job, marital, education	Integer / Categorical	Customer life-stage and socioeconomic profiling segmentation.
Financial Health	default, balance, housing, loan	Binary / Continuous (EUR)	Liquidity buffers and existing debt obligation encumbrance.
Current Contact	contact, day, month, duration, campaign	Categorical / Integer	Outbound channel, seasonality, interaction length, contact intensity.
Historical Touch	pdays, previous, poutcome	Integer / Categorical	Recency of previous contact and prior institutional response record.
Target Metric	y → subscribed	Binary (0 = No, 1 = Yes)	Ground truth term deposit subscription indicator (Target).

Data Cleaning & Feature Engineering Transformations: Column strings were lower-cased and stripped of whitespace. String literal `"unknown"` tokens were standardized to `np.nan` across categorical features to enable rigorous null tracking before recoding. Numeric features (`age, balance, day, duration, campaign, pdays, previous`) were coerced to numeric types with missing values imputed using robust median statistics. The target flag `y` was mapped into an integer indicator `subscribed` $\in \{0, 1\}$. Engineered features include five age brackets (`18-25, 26-35, 36-45, 46-55, 56+`), a prior touchpoint indicator (`contacted_before = pdays != 999`), and a campaign intensity classifier (≤3 contacts vs. >3 contacts).

Relational Data Flow & Preprocessing Pipeline



5. Implementation, Key Metrics & Descriptive Statistics

Execution was orchestrated in Python using `pandas`, `numpy`, `scipy.stats`, `statsmodels`, `matplotlib`, `seaborn`, and `sqlite3`. Distributional checks revealed intense positive skewness across balance (8.36), call duration (3.14), and campaign contacts (4.90).

CUSTOMER BASE
45,211
Verified contact records evaluated.

AVERAGE BALANCE
€1,362.27
Median: €448.00 (Std: €3,044.77)

AVG. CALL DURATION
258.16 s
Subscribers: 537.29s vs. 221.18s

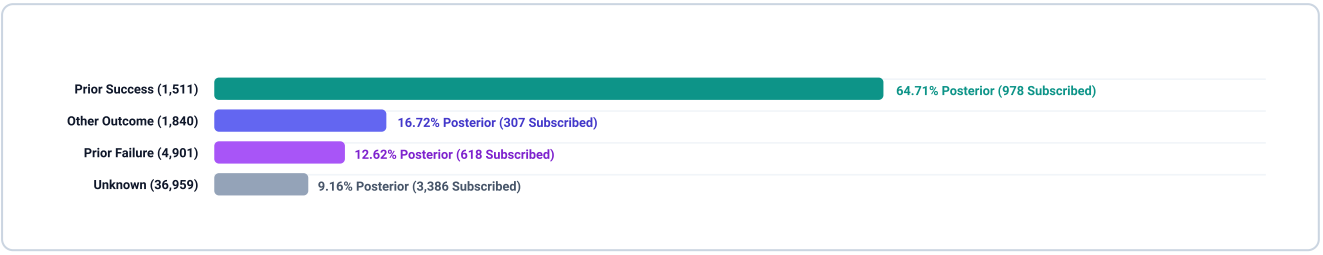
CONVERSION RATE
11.70%
5,289 subscribers enrolled.

6. Statistical Hypothesis Testing & Bayesian Updating

Three statistical tests were executed to validate that observed differences were not artifacts of random sampling:

Statistical Test Framework	Null Hypothesis (H_0)	Calculated Value & p-Value	Inference & Significance
Welch Independent t-Test	$\mu_{\text{balance, sub}} = \mu_{\text{balance, non}}$	$t = 11.23$, $p = 4.38 \times 10^{-23}$	**Reject H_0**: Subscribers hold higher liquidity buffer.
Pearson Chi-Square (χ^2)	Subscription is independent of contact type	$\chi^2 = 1035.7$, $p = 1.25 \times 10^{-225}$	**Reject H_0**: Contact medium dictates success.
A/B Two-Proportion Z-Test	$P(\text{Sub}\|\text{Cellular}) = P(\text{Sub}\|\text{Telephone})$	$Z = 2.17$, $p = 0.02996$	**Reject H_0**: Cellular conversion (14.92%) > Tel (13.42%).
Shapiro-Wilk Normality	Sample balance is normally distributed	Statistic $= 0.528$, $p = 1.68 \times 10^{-79}$	**Non-Normal**: Heavy tail justifies CLT sampling simulation.

Prior-to-Posterior Transition: Historical Campaign Impact (`poutcome`)



Using an uninformative Beta(1,1) prior, clients with previous successful campaign history jump from the 11.70% baseline to a **64.71% posterior probability**. Even clients who previously rejected an offer convert at 12.62%, outperforming completely unknown prospects (9.16%), proving that previous relationship engagement is the single strongest signal.

7. Relational SQL Insights & Aggregates

The cleaned dataset was loaded into an internal SQLite database (`bank_marketing.db`). Structured SQL queries were formulated to answer fundamental revenue-impact questions:

Analytical Dimension	SQL Core Logic	Empirical Result	Managerial Takeaway
Highest Conversion by Job	`GROUP BY job ORDER BY rate DESC`	Student (28.68%), Retired (22.79%)	Target liquidity-rich seniors and unencumbered students.
Volume Inefficiency Job	`WHERE job = 'blue-collar'`	9,732 calls → 708 subs (7.27%)	Largest staff time expenditure yields lowest success.
Top Conversion Months	`GROUP BY month ORDER BY subs DESC`	May (925), Aug (688), Jul (627)	Peak total subscriber counts cluster in late spring/summer.
Campaign Diminishing Returns	`WHERE campaign > 3`	9,641 calls → 709 subs (7.35%)	Success drops 41.5% compared to ≤3 contacts (12.56%).
Call Duration Association	`GROUP BY y AVG(duration)`	537.29s (Yes) vs. 221.18s (No)	2.4× longer engagement strongly signals conversion.
Education Premium	`GROUP BY education ORDER BY rate`	Tertiary (15.01%) vs Primary (8.63%)	Higher financial literacy correlates with deposit uptake.

8. Project Conclusion & Strategic Synthesis

The project successfully bridged raw campaign records and institutional execution. Statistical inference confirms that client balance ($p < 0.001$) and mobile contact channels ($p = 0.030$) provide statistically significant conversion advantages. The empirical distribution demonstrates that call duration is strongly coupled with positive outcomes (537s vs 221s); however, because duration is recorded during call execution, it serves as an in-flight diagnostic rather than an ex-ante targeting filter.

Critically, SQL queries exposed that contact fatigue is severe: calling a prospect more than three times drops conversion from 12.56% to 7.35%. Simultaneously, targeting 9,732 blue-collar prospects consumed 21.5% of total call volume while yielding only 7.27% conversion. Re-routing outbound resources toward previous campaign successes (64.71% posterior), students (28.68%), and retirees (22.79%) via cellular channels directly resolves marketing operational drag.

9. Future Scope & Engineering Enhancements

Future enhancements include developing an automated LightGBM / XGBoost propensity scoring model deployed as an asynchronous FastAPI inference service. This will compute real-time lead prioritization before agents dial. Additionally, multi-armed contextual bandit algorithms can be implemented to dynamically route outbound calling quotas across months, avoiding static seasonal batching and maximizing conversion velocity.

> **Professional Closing Statement: Direct marketing efficiency is fundamentally an optimization problem of statistical targeting over brute outreach. By integrating rigorous hypothesis testing, Bayesian inference, and relational query audits, commercial banking operations can systematically elevate conversion rates while curtailing client contact fatigue.**